PROBLEM 1.8

Simple Pendulum Equation of Motion

Original statement · ProblemsCh1.pdf, p. 2

Newtonian Polar Frame • Mechanical Energy • Nonlinear ODE

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** Consider a simple pendulum consisting of a bob of mass m at the end of a string of length l and negligible mass, and denote by $\theta(t)$ the angle with respect to the vertical as a function of time t. Neglecting friction, apply Newton's second law to write a differential equation for $\theta(t)$.

PHASE 2: FUNDAMENTAL FORMULATION

We present two independent methods: (1) Direct force projection in polar coordinates $(\hat{e}_r, \hat{e}_\theta)$ using Newton's second law; (2) Time differentiation of total mechanical energy $\frac{dE}{dt} = 0$.

PHASE 3: STEP-BY-STEP DERIVATION

Method 1: Polar Coordinate Force Balance:

Bob position: $\vec{r} = l\hat{e}_r$. Taking time derivatives with $\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta$ and $\dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r$:

$$\vec{v} = l\dot{\theta}\hat{e}_\theta$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -l\dot{\theta}^2\hat{e}_r + l\ddot{\theta}\hat{e}_\theta$$

The tangential acceleration is $a_\theta = l\ddot{\theta}$.

Acting forces: String tension $\vec{T} = -T\hat{e}_r$ acts purely radially. Gravity resolves as:

$$\vec{F}_g = mg\cos\theta\,\hat{e}_r - mg\sin\theta\,\hat{e}_\theta$$

Projecting Newton's second law along the tangential unit vector $\hat{e}_\theta$:

$$\sum F_\theta = ma_\theta \implies -mg\sin\theta = m(l\ddot{\theta})$$

Dividing by $ml \neq 0$ and bringing all terms to the left-hand side:

$$\ddot{\theta} + \frac{\mathbf{g}}{\mathbf{l}}\sin\theta = \mathbf{0} \tag{1.8.1}$$

Method 2: Energy Conservation Check:

$$E = E_k + E_p = \frac{1}{2}mv^2 + mgl(1-\cos\theta) = \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1-\cos\theta)$$

Differentiating with respect to time t ($\frac{dE}{dt} = 0$):

$$\frac{dE}{dt} = ml^2\dot{\theta}\ddot{\theta} + mgl\sin\theta\,\dot{\theta} = \dot{\theta}\left[ml^2\ddot{\theta} + mgl\sin\theta\right] = 0$$

For non-trivial motion ($\dot{\theta} \not\equiv 0$), dividing by ml^2 yields identically: $\ddot{\theta} + \frac{\mathbf{g}}{\mathbf{l}}\sin\theta = \mathbf{0}$.

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$\ddot{\theta}(t) + \frac{g}{l}\sin\theta(t) = 0$$

Classification: 2nd-order, autonomous, **nonlinear** ODE due to the transcendental term $\sin\theta$. For small amplitudes ($\theta \ll 1$, $\sin\theta \approx \theta$), it linearizes to the harmonic oscillator $\ddot{\theta} + \omega_0^2\theta = 0$ with natural frequency $\omega_0 = \sqrt{g/l}$.

NONLINEARITY BY THE SUPERPOSITION TEST OF THE CLASS NOTES [NOTES, P. 2]

If θ_1, θ_2 are solutions and $\theta = \alpha\theta_1 + \beta\theta_2$, then $\ddot{\theta} = -\frac{g}{l}(\alpha \sin\theta_1 + \beta \sin\theta_2)$, so $\ddot{\theta} + \frac{g}{l}\sin\theta = \frac{g}{l}\big[\sin(\alpha\theta_1 + \beta\theta_2) - \alpha\sin\theta_1 - \beta\sin\theta_2\big]$. With $\alpha = 2, \beta = 0$ this is $\frac{g}{l}(\sin 2\theta_1 - 2\sin\theta_1) = \frac{2g}{l}\sin\theta_1(\cos\theta_1 - 1)$, not zero while $\sin\theta_1 \neq 0$ and $\cos\theta_1 \neq 1$: the equation is nonlinear.